

Multilinear Formula Lower Bounds for Sparse Determinants

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Boyapati V S S Pruthvi

2024701008

boyapati.pruthvi@research.iiit.ac.in

Advisor: Dr. Suryajith Chillara



International Institute of Information Technology Hyderabad
500 032, India

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International Institute of Information Technology Hyderabad
Hyderabad, India

CERTIFICATE

This is to certify that the work presented in this thesis titled *Multilinear Formula Lower Bounds for Sparse Determinants* by Boyapati V S S Pruthvi has been carried out under my supervision and is not submitted elsewhere for a degree.

Date

Advisor: Dr. Suryajith Chillara

Abstract

Raz [Raz09] proved that multilinear formulas computing the determinant of a generic $n \times n$ matrix must have size $n^{\Omega(\log n)}$. A natural question this raises is: which properties of the determinant are actually driving this hardness? The proof simultaneously exploits several features of the polynomial (the quadratic number of variables, dense connectivity of the underlying matrix, and various symmetries), and it is not apparent which of these are genuinely necessary.

This thesis establishes that *density is not necessary*. We prove the existence of $n \times n$ symbolic matrices with only $\Theta(n \log^6 n)$ nonzero entries, a reduction in variable count by a factor of $n / \log^6 n$ relative to the full matrix, such that any multilinear formula computing their determinant still requires size $n^{\Omega(\log n)}$. The construction is based on rectangle sampling from the complete bipartite graph $K_{n,n}$. The resulting sparse graph G simultaneously possesses a perfect matching (guaranteeing that the determinant is nonzero) and exhibits *diagonal imbalance* under random vertex permutations, a geometric property that turns out to be the engine behind factor imbalance in Raz’s partial derivatives method.

The conclusion is that Raz’s technique extends to the sparse setting with no loss in the quality of the lower bound, and that the key driver of multilinear hardness for the determinant is an expansion-like combinatorial property of the variable set, not its size. The analysis relies on concentration inequalities for dependent random variables, in particular the read- k Chernoff bound due to Gavinsky, Lovett, Saks, and Srinivasan [GLSS15]. Every result stated for the determinant holds equally for the permanent.

Contents

Abstract	iii
1 Introduction	1
1.1 Background	1
1.2 Main Results	3
1.3 Proof Overview	3
1.4 Organisation	6
2 Preliminaries	7
2.1 Edmonds' Criterion	7
2.2 Algebraic Complexity Classes	8
2.3 Raz's Partial Derivatives Measure	9
2.4 Imbalance and Rank Deficiency	9
2.5 Concentration Inequalities	10
3 Sparse Matrix Construction via Rectangle Sampling	12
3.1 The Sampling Process	12
3.2 Edge Survival Probabilities	13
3.3 Diagonal Imbalance Under Random Permutations	14
3.4 Verification of Hall's Marriage Condition	15
3.4.1 A uniform count	15
3.4.2 Small sets	15
3.4.3 Moderate sets	16
3.4.4 Near-full sets	17
3.5 The Sparse Matrix	17
4 Lower Bounds Against Multilinear Formulas	19
4.1 Second-Round Sampling and Concentration	19
4.2 The Random Restriction	21
4.3 Balance Analysis	21
4.4 Read-2 Structure of the Balance Events	22

4.5	Many Factors are Simultaneously Imbalanced	22
4.6	The Lower Bound	23
5	Discussion	25

Chapter 1

Introduction

Two polynomials have dominated algebraic complexity theory since Valiant’s early work in the late 1970s: the determinant and the permanent of an $n \times n$ symbolic matrix. They behave quite differently. The permanent is **VNP**-complete [Val79a] and computing it over $\{0, 1\}$ -matrices is **#P**-complete [Val79b], placing it firmly among the hardest polynomials we know. The determinant sits at the other end: it lies in **VP** and even characterises **VP** under quasi-polynomial projections [Val79a]. The gap between the two is widely viewed as the algebraic analogue of the **P** versus **NP** question.

Despite the determinant’s relative simplicity compared to the permanent, obtaining lower bounds against restricted models has proved harder than one might expect. The strongest such result is due to Raz [Raz04, Raz09]: any syntactically multilinear formula computing the $n \times n$ determinant requires size $n^{\Omega(\log n)}$. This thesis asks a natural follow-up: which properties of the determinant actually drive this hardness? We show that density, commonly viewed as a defining feature of the determinant, is not one of them.

1.1 Background

Determinant as an iterated matrix product. Over a field, Gaussian elimination computes the determinant in polynomial time, but this algorithm is inherently sequential and breaks down for general rings or in the parallel setting. Berkowitz [Ber84] gave an algorithm that computes the characteristic polynomial of any $n \times n$ matrix (and hence the determinant) in $O(\log^2 n)$ parallel time using polynomially many processors, by exploiting iterated matrix multiplication and the Cayley-Hamilton theorem. Mahajan and Vinay [MV97] gave a different perspective via closed walk sequences (clow sequences), leading to an ABP-based computation that places the determinant in the complexity class **GapL**. Both al-

gorithms arrive at the same point: the determinant, despite appearing sequential on the surface, can be expressed as an iterated matrix product. This is precisely what it means for a polynomial to lie in the class VBP.

Multilinear formulas. A polynomial is *multilinear* when every variable appears to degree at most one in each monomial. Both the determinant and the permanent are multilinear. A formula is *syntactically multilinear* when this constraint is enforced gate by gate: every gate must compute a multilinear polynomial. Classical algorithms for the determinant (Berkowitz, Mahajan-Vinay) involve non-multilinear intermediate expressions that cancel out at the output. Restricting to syntactically multilinear formulas forbids such cancellations, and one would expect this to make the computation harder.

Indeed it does. Raz [Raz04, Raz09] proved that any multilinear formula for the determinant or permanent of an $n \times n$ matrix has size at least $n^{\Omega(\log n)}$. This is the strongest lower bound known for an explicit polynomial in the formula model. Subsequently, Raz [Raz06] showed that multilinear circuits are superpolynomially more powerful than multilinear formulas; a streamlined exposition appears in [RY09].

Further developments. Much further work has followed. Raz and Yehudayoff [RY09] conjectured that multilinear ABPs are exponentially stronger than multilinear formulas. This was confirmed by Dvir, Malod, Perifel, and Yehudayoff [DMPY16], who exhibited an explicit polynomial family in multilinear-ABP that requires superpolynomial multilinear formula size. Raz and Yehudayoff [RY09] also established a depth hierarchy for multilinear formulas: for each constant k , there exist polynomials computable by depth- $(k + 1)$ multilinear formulas that provably require depth- k formulas of superpolynomial size. Chillara, Engels, Limaye, and Srinivasan [CELS18] subsequently made this hierarchy near-optimal.

An open problem raised by Shpilka and Yehudayoff [SY10, Open Problem 14] asks whether general formulas are strictly more powerful than multilinear ones. Chillara, Limaye, and Srinivasan [CLS19] showed, over large fields, that general formulas outperform multilinear formulas of depth $o(\log n)$. Resolving the question at depth $O(\log n)$ would be enough, since the depth reduction of formulas to $O(\log n)$ [BKM73, Bre74] is tight [CLS19].

1.2 Main Results

Raz’s proof [Raz04, Raz09] uses several properties of the determinant simultaneously: the polynomial involves n^2 variables arranged in a fully connected $n \times n$ matrix, it has rich permutation-based structure, and its monomials are indexed by permutations. The proof does not easily decompose into independent contributions from each of these features, so it is not clear which ones are load-bearing.

Our investigation focuses on density. The determinant of a generic $n \times n$ matrix involves n^2 variables. What happens when most of these variables are set to zero, leaving a sparse matrix? If the lower bound persists, density was not the point.

Theorem 1.1 (Lower Bound for Sparse Determinants). *There exists a family of $n \times n$ sparse symbolic matrices $(X_G)_{n \in \mathbb{N}}$ constructed by probabilistic sampling with the following properties:*

1. *Each row and column of X_G has exactly $\Theta(\log^6 n)$ nonzero entries.*
2. *The bipartite graph G underlying X_G has a perfect matching, so $\text{Det}(X_G) \neq 0$.*
3. *Any multilinear formula computing $\text{Det}(X_G)$ has size at least $n^{\Omega(\log n)}$.*

The variable count drops from n^2 to $\Theta(n \log^6 n)$, a reduction by a factor of $n / \log^6 n$, yet the lower bound $n^{\Omega(\log n)}$ is unchanged. This is the best exponent one could hope for in Raz’s framework.

Remark 1.2 Every result stated for the determinant holds equally for the permanent.

1.3 Proof Overview

The argument builds on Raz’s partial derivatives method [Raz04], adapting it to the sparse setting. We first recall the method and then describe where our proof diverges from Raz’s.

Raz’s partial derivatives measure. Fix a multilinear polynomial f over variable set X . Given an equi-partition $\sigma : X \mapsto Y \sqcup Z$ with $|Y| = |Z| = |X|/2$, form the coefficient matrix M_f whose rows are indexed by multilinear monomials over Y and whose columns are indexed by multilinear monomials over Z , with entry (m_Y, m_Z) equal to the coefficient of $m_Y \cdot m_Z$ in f . Raz’s complexity measure of f under σ is $\mu_\sigma(f) = \text{rank}(M_f)$, and the trivial upper bound gives $\mu_\sigma(f) \leq 2^{|X|/2}$.

What matters is how μ_σ behaves on products. If $f = f_1 \times f_2$ with f_1, f_2 defined over disjoint sets X_1, X_2 , then $M_f = M_{f_1} \otimes M_{f_2}$, so $\mu_\sigma(f) = \mu_\sigma(f_1) \cdot \mu_\sigma(f_2)$. Now

if σ assigns more variables of X_1 to Y than to Z (say $||Y_1| - |Z_1|| \geq 2k$), then the coefficient matrix of f_1 is very rectangular, and $\mu_\sigma(f_1) \leq 2^{(|X_1| - 2k)/2} = 2^{|X_1|/2} \cdot 2^{-k}$. In general, if a product of t factors has at least ℓ factors with this “ k -imbalance” property, then $\mu_\sigma(f) \leq 2^{|X|/2} \cdot 2^{-\ell k}$, an exponential rank deficiency.

Every multilinear formula F of size s computing a polynomial f over N variables admits a sum-of-products decomposition (due to [SY10], rephrasing ideas of Raz [Raz04])

$$f = \sum_{i=1}^{s+1} f_i, \quad f_i = \prod_{j=1}^t f_{ij},$$

where $t = \Theta(\log N)$ and the variable set X_{ij} of each factor f_{ij} satisfies $(\frac{1}{3})^j N \leq |X_{ij}| \leq (\frac{2}{3})^j N$. In particular, $\Theta(\log N)$ of these factors contain at least N^δ variables for any fixed $\delta < 1$. To prove a lower bound, one therefore needs a random partition σ for which:

1. The polynomial of interest achieves high rank, and
2. Every summand in the formula decomposition has exponentially smaller rank.

Applying the measure directly to the n^2 -variable determinant does not work: the matrix M_f would have $2^{n^2/2}$ rows and columns, far exceeding the number of monomials in the determinant, so full rank is impossible. Raz’s fix [Raz04] is to embed m disjoint 2×2 blocks along a generalised diagonal (setting all other entries to zero or one) and sample this embedding at random. The restricted determinant becomes $\prod_{i=1}^m (y_i - z_i)$, a polynomial of full rank 2^m over its $2m$ active variables. On the formula side, each factor in the decomposition becomes imbalanced under this restriction with good probability, yielding the bound.

Our construction. Rather than working with the dense matrix and restricting, we start with a sparse matrix X_G whose nonzero support is defined by a randomly sampled bipartite graph. The graph G is built via *rectangle sampling*: each of the $N = \binom{n}{2}^2$ possible 2×2 submatrices (rectangles) of the $n \times n$ matrix is included independently with probability $p = \log^6 n / n^3$, and an entry (i, j) is kept as a variable if and only if it appears in at least one sampled rectangle. This yields a sparse graph G with $\Theta(n \log^6 n)$ edges and degree $\Theta(\log^6 n)$ per vertex.

Two things are needed from G : (a) a perfect matching (so that the determinant is nonzero), and (b) a geometric property called *diagonal imbalance*.

For (a), we verify Hall’s marriage condition directly, analysing small, moderate, and near-full subsets separately (Theorems 3.7, 3.9 and 3.10, combined in Theorem 3.11). We further show that a perfect matching survives after removing

any $2m = \Theta(n^{1/3})$ rows and columns (Theorem 3.12), which is needed for the restriction step.

For (b): given a random permutation $\sigma = (\sigma_L, \sigma_R)$ of the vertices, each rectangle acquires an orientation (top left, top right, bottom left, bottom right corners). For a set S of edges with $|S| \geq |E_G|^{0.75}$, say that a rectangle is *diagonally imbalanced relative to S* if exactly one of its two opposite corners (top left and bottom right, or top right and bottom left) belongs to S . Theorem 3.5 shows that with high probability over σ , at least $(1 - \delta)|S|/2$ of the rectangles in $\mathcal{R}$ are diagonally imbalanced relative to S , for any constant $\delta > 0$. This is proved via McDiarmid's inequality: changing one coordinate of σ affects at most $O(\log^6 n)$ rectangles (the maximum degree), so the count is well concentrated.

The random restriction. We sample $m = \Theta(n^{1/3})$ rectangles from $\mathcal{R}$ (with no two sharing a row or column), and for each sampled rectangle R_i we independently choose one of two configurations: either the top left and bottom right corners become active variables (y_i, z_i) and the other diagonal is set to 1, or vice versa. The remaining variables are assigned 0 or 1 according to a fixed perfect matching in the residual graph. This creates an equi-partition $X \mapsto Y \sqcup Z$ with $|Y| = |Z| = m$, which is the partition fed into Raz's measure.

The central technical fact (Theorem 4.4) is that for any factor f_{ij} in the formula decomposition with $|X_{ij}| \geq N^{0.75}$, the number of sampled rectangles contributing diagonal imbalance to f_{ij} is at least $\Omega(n^{1/12} \log^{-1.5} n)$ with high probability. Since each such rectangle assigns one variable to Y_{ij} and one to Z_{ij} independently with probability $1/2$ each, the count $|Y_{ij}|$ follows an approximate binomial distribution. The probability that f_{ij} is k -balanced (meaning $||Y_{ij}| - |Z_{ij}|| < 2k$ for $k = n^{1/32}$) is then at most $O(n^{-1/96} \log^{0.75} n)$ (Theorem 4.7).

Concluding the bound. Conditioned on the set of sampled rectangles, the balance events for the different factors form a *read-2 family* with respect to the configuration choices (Theorem 4.8): each rectangle's configuration affects at most two factors, since its two active variables each belong to at most one factor in the product. Applying the read-2 Chernoff bound of [GLSS15] (Theorem 2.15) shows that with probability at least $1 - \exp(-\Omega(\log^2 n))$, at least half the large factors in any single summand are simultaneously k -imbalanced (Theorem 4.9).

By Theorem 2.10, each summand then has rank at most $2^m \cdot 2^{-\Omega(n^{1/32} \log n)}$. A union bound over all $s + 1$ summands shows that if $s = n^{O(\log n)}$, the formula achieves total rank $(s + 1) \cdot 2^m \cdot 2^{-\Omega(n^{1/32} \log n)} \ll 2^m$ with probability $1 - o(1)$. But the restricted determinant expands as $\prod_{i=1}^m (y_i z_i - 1)$, which has coefficient matrix of full rank 2^m (the residual graph retains a perfect matching by Theorem 3.12).

This contradiction gives $s = n^{\Omega(\log n)}$.

1.4 Organisation

Chapter 2 covers the algebraic complexity background and the concentration inequalities used throughout. Chapter 3 constructs the sparse matrix X_G and establishes its two key properties. Chapter 4 carries out the lower bound argument. Chapter 5 examines the design choices in the construction and what they suggest about the nature of multilinear hardness.

Chapter 2

Preliminaries

We write $[n]$ for the set $\{1, 2, \dots, n\}$. Throughout, X denotes an $n \times n$ symbolic matrix whose entry in row i and column j is a distinct formal variable $x_{i,j}$. We identify X with the complete bipartite graph $K_{n,n}$, where the left vertex u_i corresponds to row i and the right vertex v_j corresponds to column j , with edge (u_i, v_j) carrying the variable $x_{i,j}$.

Definition 2.1 (Rectangle). *A rectangle is a 2×2 submatrix of X . Concretely, rows $i < i'$ and columns $j < j'$ determine the rectangle $R = \{(i, j), (i, j'), (i', j), (i', j')\}$. The complete bipartite graph $K_{n,n}$ contains exactly $N = \binom{n}{2}^2$ rectangles.*

Definition 2.2 (Corner Positions). *Fix a rectangle with rows $\{i, i'\}$ ($i < i'$) and columns $\{j, j'\}$ ($j < j'$), together with an ordering $\sigma = (\sigma_L, \sigma_R)$ of the left and right vertices. The top left corner is (i, j) , the top right is (i, j') , the bottom left is (i', j) , and the bottom right is (i', j') .*

2.1 Edmonds' Criterion

Whether the determinant of a symbolic matrix is identically zero as a polynomial is decided purely by the support graph of the matrix.

Definition 2.3 (Edmonds Graph). *The Edmonds graph of an $n \times n$ symbolic matrix $A = (x_{i,j})$ is the bipartite graph $G_A = (V_1 \sqcup V_2, E)$ with $V_1 = \{u_1, \dots, u_n\}$ and $V_2 = \{v_1, \dots, v_n\}$, where $(u_i, v_j) \in E$ precisely when $x_{i,j} \neq 0$.*

Lemma 2.4 (Edmonds' Criterion). *For an $n \times n$ symbolic matrix A with Edmonds graph G_A , we have $\text{Det}(A) \neq 0$ as a polynomial if and only if G_A has a perfect matching.*

Proof. By definition, $\text{Det}(A) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{i=1}^n x_{i, \sigma(i)}$. Since the $x_{i,j}$ are independent formal variables, this sum is nonzero if and only if at least one term is

nonzero. The term corresponding to σ is nonzero precisely when every $x_{i,\sigma(i)}$ is a variable (not identically zero), i.e., when the set of edges $\{(u_i, v_{\sigma(i)}) : i \in [n]\}$ is a perfect matching in G_A . $\square$

2.2 Algebraic Complexity Classes

An *arithmetic circuit* over a field $\mathbb{F}$ is a directed acyclic graph whose leaves are labelled by variables or constants in $\mathbb{F}$, and whose internal nodes are labelled by $+$ or $\times$. Every gate computes a polynomial in $\mathbb{F}[x_1, \dots, x_n]$ in the natural way. The *size* of a circuit is its number of edges.

Definition 2.5 (VF, VBP, VP, VNP). *Let $(f_n)_{n \in \mathbb{N}}$ be a polynomial family where $f_n \in \mathbb{F}[x_1, \dots, x_n]$ has degree at most $\text{poly}(n)$.*

- $(f_n) \in \text{VF}$ if f_n is computed by an arithmetic formula (a circuit of tree shape, where every gate has fanout one) of size $\text{poly}(n)$.
- $(f_n) \in \text{VBP}$ if f_n is computed by an algebraic branching program (ABP) of size $\text{poly}(n)$. An ABP is a layered directed acyclic graph with a single source and a single sink; each edge carries a linear polynomial as its weight, and the program evaluates to the sum over all source-to-sink paths of the product of edge weights along each path.
- $(f_n) \in \text{VP}$ if f_n is computed by an arithmetic circuit of size $\text{poly}(n)$ and degree $\text{poly}(n)$.
- $(f_n) \in \text{VNP}$ if there exists a family $(g_n) \in \text{VP}$ with $m = \text{poly}(n)$ such that

$$f_n(x_1, \dots, x_n) = \sum_{e \in \{0,1\}^m} g_{n+m}(x_1, \dots, x_n, e_1, \dots, e_m).$$

The known inclusions among these classes are

$$\text{VF} \subseteq \text{VBP} \subseteq \text{VP} \subseteq \text{VNP},$$

mirroring the Boolean containments $\text{NC}^1 \subseteq \text{L} \subseteq \text{P} \subseteq \text{NP}$. All four containments are conjectured to be strict, but none has been proved.

Several facts about these classes are relevant to our work. The determinant of an $n \times n$ matrix lies in VBP and moreover characterises VP under quasi-polynomial projections [Val79a]. The permanent is VNP-complete under p-projections [Val79a], and computing the permanent of a $\{0, 1\}$ -matrix is $\#\text{P}$ -hard over the integers [Val79b].

A further restriction of interest is to *multilinear* computation. A polynomial is multilinear when every variable appears to degree at most one in each monomial. An arithmetic circuit is *syntactically multilinear* when every gate computes a multilinear polynomial. This is a strong restriction: the standard polynomial-time algorithms for the determinant (Berkowitz [Ber84], Mahajan-Vinay [MV97]) pass through non-multilinear intermediate computations that cancel at the output. Raz [Raz04, Raz09] showed that no syntactically multilinear formula of polynomial size exists for the determinant or permanent; any such formula must have size $n^{\Omega(\log n)}$.

It is not known whether the determinant lies in the multilinear analogue of VBP. Dvir, Malod, Perifel, and Yehudayoff [DMPY16] separated multilinear formulas from multilinear ABPs: there is an explicit polynomial in multilinear-VBP that requires superpolynomial multilinear formula size.

2.3 Raz's Partial Derivatives Measure

Definition 2.6 (Coefficient Matrix). *Let f be a multilinear polynomial over X , and let $\sigma : X \rightarrow Y \sqcup Z$ be an equi-partition with $|Y| = |Z| = |X|/2$. Let $\mathcal{M}_Y$ (resp. $\mathcal{M}_Z$) denote the set of all multilinear monomials over $\sigma^{-1}(Y)$ (resp. $\sigma^{-1}(Z)$). The coefficient matrix M_f^σ has rows indexed by $\mathcal{M}_Y$ and columns by $\mathcal{M}_Z$, with entry (m_Y, m_Z) equal to the coefficient of $m_Y \cdot m_Z$ in f . We write $\mu_\sigma(f) = \text{rank}(M_f^\sigma)$.*

Observation 2.7. *For any equi-partition σ of a multilinear polynomial f over $|X|$ variables, $\mu_\sigma(f) \leq 2^{|X|/2}$.*

Proposition 2.8 (Multiplicativity). *If $f = f_1 \times f_2$ with f_1, f_2 multilinear over disjoint sets X_1, X_2 and $X = X_1 \sqcup X_2$, then*

$$M_f^\sigma = M_{f_1}^{\sigma|_{X_1}} \otimes M_{f_2}^{\sigma|_{X_2}} \quad \text{and} \quad \mu_\sigma(f) = \mu_{\sigma|_{X_1}}(f_1) \cdot \mu_{\sigma|_{X_2}}(f_2).$$

Proof. Each monomial of f factors uniquely as $m_1 \cdot m_2$ with m_i a monomial of f_i . The restricted partition $\sigma|_{X_i}$ induces the corresponding row/column split on $M_{f_i}^{\sigma|_{X_i}}$. The coefficient of (m_Y, m_Z) in f factors as the product of the $(m_Y \cap m_1, m_Z \cap m_1)$ coefficient of f_1 and the $(m_Y \cap m_2, m_Z \cap m_2)$ coefficient of f_2 , so M_f^σ is the Kronecker product, giving the rank formula. $\square$

2.4 Imbalance and Rank Deficiency

Definition 2.9 (k -Imbalance). *Given a multilinear polynomial f over X and an equi-partition $\sigma : X \rightarrow Y \sqcup Z$, say f is k -imbalanced under σ if $||Y| - |Z|| \geq 2k$,*

and k -balanced if $||Y| - |Z|| < 2k$.

Proposition 2.10 (Rank Loss). *If f is multilinear over $|X|$ variables and k -imbalanced under σ , then $\mu_\sigma(f) \leq 2^{|X|/2} \cdot 2^{-k}$. More generally, if $f = \prod_{i=1}^t f_i$ with the f_i over pairwise disjoint variable sets and at least ℓ factors are k -imbalanced, then $\mu_\sigma(f) \leq 2^{|X|/2} \cdot 2^{-\ell k}$.*

Proof. For a single k -imbalanced factor: assume $|Y| \leq |Z|$, so $|Y| = (|X| - \Delta)/2$ and $|Z| = (|X| + \Delta)/2$ with $\Delta \geq 2k$. Then M_f^σ has at most $2^{|Y|}$ rows, giving $\mu_\sigma(f) \leq 2^{|Y|} \leq 2^{(|X|-2k)/2}$. For the product: by Theorem 2.8, $\mu_\sigma(f) = \prod_i \mu_{\sigma|_{X_i}}(f_i)$; each imbalanced factor contributes at most $2^{|X_i|/2-k}$ instead of $2^{|X_i|/2}$, giving a deficit of 2^{-k} per imbalanced factor. $\square$

The reason Theorem 2.10 is useful is the following formula decomposition theorem, which converts any small multilinear formula into a sum of products with a controlled variable structure.

Theorem 2.11 (Formula Decomposition [SY10, Sap21]). *Every multilinear formula F of size s over N variables admits a decomposition*

$$f = \sum_{i=1}^{s+1} f_i, \quad f_i = \prod_{j=1}^t f_{ij},$$

where $t = \Theta(\log N)$ and the variable set X_{ij} of each factor satisfies $(\frac{1}{3})^j N \leq |X_{ij}| \leq (\frac{2}{3})^j N$ for each level $j \in [t]$.

2.5 Concentration Inequalities

The following three results are used in the proofs of Chapters 3 and 4.

Theorem 2.12 (Chernoff Bound [MU17]). *Let $X_1, \dots, X_n$ be independent $\{0, 1\}$ -valued random variables with $\Pr[X_i = 1] = p_i$. Set $X = \sum_{i=1}^n X_i$ and $\mu = \mathbb{E}[X]$. For any $\delta \in (0, 1]$,*

$$\Pr[X \geq (1 + \delta)\mu] \leq \exp\left(-\frac{\delta^2 \mu}{3}\right), \quad \Pr[X \leq (1 - \delta)\mu] \leq \exp\left(-\frac{\delta^2 \mu}{2}\right).$$

Theorem 2.13 (McDiarmid's Inequality [McD89]). *Let $\mathcal{X}$ be a set, and let $X_1, \dots, X_n$ be independent $\mathcal{X}$ -valued random variables. Let $f : \mathcal{X}^n \rightarrow \mathbb{R}$ have the bounded differences property: there exist constants $c_1, \dots, c_n > 0$ such that for every $i \in [n]$ and every $x_1, \dots, x_n, x'_i \in \mathcal{X}$,*

$$|f(x_1, \dots, x_i, \dots, x_n) - f(x_1, \dots, x'_i, \dots, x_n)| \leq c_i.$$

Then for any $t > 0$,

$$\Pr[|f(X_1, \dots, X_n) - \mathbb{E}[f(X_1, \dots, X_n)]| \geq t] \leq 2 \exp\left(-\frac{2t^2}{\sum_{i=1}^n c_i^2}\right).$$

Definition 2.14 (Read- k Family). *Let $Y_1, \dots, Y_r$ be events determined by independent random variables $X_1, \dots, X_m$. The collection $\{Y_i\}$ forms a read- k family if each X_j influences at most k of the events Y_i .*

Theorem 2.15 ([GLSS15]). *Let $Y_1, \dots, Y_r$ be a read- k family of indicator variables with $\Pr[Y_i = 1] = p_i$. Set*

$$p = \frac{1}{r} \sum_{i=1}^r p_i.$$

For any $\varepsilon > 0$,

$$\mathbb{P}[Y_1 + \dots + Y_r \geq (p + \varepsilon)r] \leq e^{-D(p+\varepsilon \| p) \cdot r/k}$$

and

$$\mathbb{P}[Y_1 + \dots + Y_r \leq (p - \varepsilon)r] \leq e^{-D(p-\varepsilon \| p) \cdot r/k},$$

where

$$D(q \| p) = q \log\left(\frac{q}{p}\right) + (1 - q) \log\left(\frac{1 - q}{1 - p}\right) \geq 2(q - p)^2$$

is the Kullback-Leibler divergence.

Chapter 3

Sparse Matrix Construction via Rectangle Sampling

The polynomial we analyse is the determinant of a randomly sampled sparse matrix. Let X be the generic $n \times n$ matrix corresponding to the complete bipartite graph $K_{n,n}$. We will thin out X by removing most of its entries (setting them to zero), obtaining a sparse matrix X_G whose nonzero support is the edge set E_G of a sparse bipartite graph G . Formally, $\text{Det}(X_G)$ is the determinant of the matrix with $(X_G)_{i,j} = x_{i,j}$ when $(i,j) \in E_G$ and $(X_G)_{i,j} = 0$ otherwise. Three properties of X_G are needed:

1. $\text{Det}(X_G) \neq 0$ as a polynomial (so the polynomial is interesting);
2. $|E_G| = \Theta(n \log^6 n)$ (sparsity);
3. diagonal imbalance under random vertex permutations (the engine of the lower bound).

The construction that achieves all three simultaneously is *rectangle sampling*.

3.1 The Sampling Process

Definition 3.1 (Rectangle Sampling). *Sample a random set $\mathcal{R}$ of rectangles by retaining each of the $N = \binom{n}{2}^2$ possible rectangles independently with probability $p = \frac{\log^6 n}{n^3}$. An edge (i,j) survives in G if and only if it belongs to at least one rectangle in $\mathcal{R}$.*

By linearity of expectation, $\mathbb{E}[|\mathcal{R}|] = N \cdot p = \binom{n}{2}^2 \cdot \frac{\log^6 n}{n^3} = \Theta(n \log^6 n)$.

Proposition 3.2 (Concentration of $|\mathcal{R}|$). *With probability at least $1 - \exp(-\Omega(n \log^6 n))$, we have $|\mathcal{R}| = \Theta(n \log^6 n)$.*

Proof. The rectangles are sampled independently, so $|\mathcal{R}|$ is a sum of N independent Bernoulli(p) random variables with mean $\mu = Np = \Theta(n \log^6 n)$. By Theorem 2.12 with $\delta = 1/2$,

$$\begin{aligned}\Pr[|\mathcal{R}| \geq \tfrac{3}{2}\mu] &\leq \exp\left(-\frac{\mu}{12}\right), \\ \Pr[|\mathcal{R}| \leq \tfrac{1}{2}\mu] &\leq \exp\left(-\frac{\mu}{8}\right).\end{aligned}$$

A union bound over both tails gives

$$\Pr[|\mathcal{R}| \in [\tfrac{1}{2}\mu, \tfrac{3}{2}\mu]] \geq 1 - 2\exp\left(-\frac{\mu}{12}\right) = 1 - \exp(-\Omega(n \log^6 n)). \quad \square$$

3.2 Edge Survival Probabilities

Each edge (i, j) appears as a corner in exactly $(n-1)^2$ rectangles (one additional row and one additional column, each chosen in $(n-1)$ ways). The probability that (i, j) survives is

$$\begin{aligned}\Pr[(i, j) \text{ survives}] &= 1 - (1-p)^{(n-1)^2} \approx 1 - e^{-(n-1)^2 p} \\ &\approx (n-1)^2 p = \frac{\log^6 n}{n} \cdot (1 - 1/n)^2 \approx \frac{\log^6 n}{n}.\end{aligned}$$

By linearity of expectation, $\mathbb{E}[|E_G|] \approx n^2 \cdot \frac{\log^6 n}{n} = n \log^6 n$.

Proposition 3.3 (Edge Concentration via Read-4 Chernoff). *With probability at least $1 - \exp(-\Omega(n \log^6 n))$, we have $|E_G| = \Theta(n \log^6 n)$. Moreover, each vertex degree is $\Theta(\log^6 n)$ with probability at least $1 - \exp(-\Omega(\log^6 n))$.*

Proof. Each edge e is present if and only if at least one of the $(n-1)^2$ rectangles containing it is in $\mathcal{R}$. The edge indicator Y_e is therefore a function of the rectangle indicators $\{X_R : e \in R\}$, and each rectangle indicator X_R influences the four edge indicators corresponding to its four corners. This is a read-4 structure (Theorem 2.14). The mean edge count is $\Theta(n \log^6 n)$, and applying Theorem 2.15 gives the stated concentration. The degree bound follows from the same argument restricted to rectangles incident to a fixed vertex, which has a read-2 structure. $\square$

For a surviving edge e , let N_e denote the number of rectangles in $\mathcal{R}$ that contain e . We have $\mathbb{E}[N_e] = (n-1)^2 \cdot p \approx \frac{\log^6 n}{n}$, and $N_e = \Theta\left(\frac{\log^6 n}{n}\right)$ with high probability. In particular, $N_e \geq 1$ whenever e survives.

3.3 Diagonal Imbalance Under Random Permutations

Given an ordering $\sigma = (\sigma_L, \sigma_R)$ of the left and right vertices, each rectangle acquires top left, top right, bottom left, and bottom right corners as in Theorem 2.2.

Definition 3.4 (Diagonal Rectangle Sets). *For a set $S \subseteq E_G$ and an ordering σ , let $\mathcal{R}_{S,\text{diag}}(\sigma)$ be the set of rectangles in $\mathcal{R}$ for which exactly one of the two pairs of opposite corners ($\{\text{top left, bottom right}\}$ or $\{\text{top right, bottom left}\}$) has exactly one member in S .*

Theorem 3.5 (Diagonal Imbalance Concentration). *Let $S \subseteq E_G$ with $|S| \geq |E_G|^{0.75}$. With probability at least $1 - 2\exp(-\Omega(n^{0.5} \log^{-3} n))$ over a uniformly random permutation $\sigma = (\sigma_L, \sigma_R)$,*

$$|\mathcal{R}_{S,\text{diag}}(\sigma)| \geq (1 - \delta) \cdot \frac{|S|}{2}$$

for any constant $\delta > 0$.

Proof. For a rectangle R with four edges $e_1 = (i, j)$, $e_2 = (i, j')$, $e_3 = (i', j)$, $e_4 = (i', j')$, define the indicator $W_R(\sigma) = 1$ if exactly one of $\{e_1, e_4\}$ or exactly one of $\{e_2, e_3\}$ is in S . Then $|\mathcal{R}_{S,\text{diag}}(\sigma)| = \sum_{R \in \mathcal{R}} W_R(\sigma)$.

Expectation. The permutation σ independently and uniformly decides which pair of opposite corners becomes the top-left/bottom-right diagonal versus the other. So $\mathbb{E}[W_R(\sigma)] = \frac{1}{2}[\mathbf{1}[e_1 \in S \oplus e_4 \in S] + \mathbf{1}[e_2 \in S \oplus e_3 \in S]]$, where $\oplus$ is XOR. Summing over rectangles and rearranging by edges, with $N_e^{(S)}$ denoting the number of rectangles in $\mathcal{R}$ where the diagonal opposite of e falls in S :

$$\mathbb{E}[|\mathcal{R}_{S,\text{diag}}(\sigma)|] = \frac{1}{2} \left[\sum_{e \in S} (N_e - N_e^{(S)}) + \sum_{e \notin S} N_e^{(S)} \right] = \frac{1}{2} \sum_{e \in S} N_e \geq \frac{|S|}{2},$$

since $N_e \geq 1$ for every surviving edge.

Concentration. Changing a single coordinate of σ alters the orientation of at most $c = O(\log^6 n)$ rectangles (the maximum vertex degree). By Theorem 2.13 applied to the $2n$ coordinates of σ , with each bounded difference at most c :

$$\Pr[||\mathcal{R}_{S,\text{diag}}(\sigma)| - \mathbb{E}[|\mathcal{R}_{S,\text{diag}}(\sigma)|]| > t] \leq 2\exp\left(-\frac{2t^2}{2nc^2}\right).$$

Setting $t = \delta \cdot \mathbb{E}[|\mathcal{R}_{S,\text{diag}}(\sigma)|] = \Theta(n^{0.75} \log^{4.5} n)$ yields

$$\Pr[\text{deviation}] \leq 2\exp\left(-\Omega\left(\frac{n^{1.5} \log^9 n}{n \log^{12} n}\right)\right) = 2\exp(-\Omega(n^{0.5} \log^{-3} n)). \quad \square$$

Remark 3.6 Theorem 3.5 captures the combinatorial heart of the construction. Any large set of edges, regardless of how it is chosen, will see roughly half its associated rectangles exhibit diagonal imbalance under a random permutation. This “adversary-proof” property is what the lower bound analysis exploits in Chapter 4.

3.4 Verification of Hall’s Marriage Condition

We now establish that the sparse graph G has a perfect matching with high probability. By Theorem 2.4, this is equivalent to $\text{Det}(X_G) \neq 0$. The tool is Hall’s theorem: G has a perfect matching if and only if $|N(S)| \geq |S|$ for every $S \subseteq V_1$, where $N(S)$ is the set of neighbours of S in V_2 .

We split the analysis into three regimes based on the size of S .

3.4.1 A uniform count

Fix $S \subseteq V_1$ with $|S| = k$ and $T \subseteq V_2$ with $|T| = \ell$. Let $M_{T,S}$ count the rectangles in $\mathcal{R}$ that are incident to at least one row in S and at least one column in T . With $C_{i,T}$ denoting the rectangles incident to row i with at least one column in T :

$$|C_{i,T}| = (n-1) \left[\binom{n}{2} - \binom{n-\ell}{2} \right], \quad |C_{i,T} \cap C_{j,T}| = \binom{n}{2} - \binom{n-\ell}{2}.$$

By inclusion-exclusion,

$$\begin{aligned} M_{T,S} &= \left| \bigcup_{i \in S} C_{i,T} \right| \geq \sum_{i \in S} |C_{i,T}| - \sum_{\substack{i,j \in S \\ i \neq j}} |C_{i,T} \cap C_{j,T}| \\ &= k \left[\binom{n}{2} - \binom{n-\ell}{2} \right] \left(n - \frac{k+1}{2} \right) =: \alpha(n, k, \ell). \end{aligned}$$

The probability that T is completely disjoint from $N(S)$ satisfies $\Pr[T \cap N(S) = \emptyset] \leq (1-p)^{M_{T,S}} \leq e^{-p\alpha(n,k,\ell)}$, so

$$\Pr[|N(S)| < k] \leq \sum_{\ell \geq n-k+1} \binom{n}{\ell} \exp(-p\alpha(n, k, \ell)).$$

3.4.2 Small sets: $|S| \leq n/\log n$

Theorem 3.7 (Hall for Small Sets). *With probability at least $1 - \exp(-\Omega(\log^6 n))$, Hall’s condition holds for every S with $1 \leq |S| \leq n/\log n$.*

Proof. Observe that $\binom{n}{2} - \binom{n-\ell}{2} = \ell(2n-\ell-1)/2$. For $\ell \geq n-k+1$ this expression is minimised at $\ell = n-k+1$, giving $\alpha(n, k, \ell) \geq k \cdot \frac{(n-k+1)(n+k-2)}{2} \cdot (n - \frac{k+1}{2})$. For

$$k \leq n/\log n,$$

$$\alpha(n, k, \ell) \geq k \left(n - \frac{n}{\log n} \right) \cdot n \cdot \left(n - \frac{n}{2 \log n} \right) \geq \frac{kn^3}{8}.$$

Hence, using $p = \log^6 n/n^3$ and the union bound over k and ℓ :

$$\begin{aligned} & \Pr \left[\exists S : |S| \leq \frac{n}{\log n}, |N(S)| < |S| \right] \\ & \leq \sum_{k=1}^{n/\log n} \binom{n}{k} \cdot k \cdot \binom{n}{n-k} \cdot \exp \left(-\frac{\log^6 n}{n^3} \cdot \frac{kn^3}{8} \right) \\ & \leq \sum_{k=1}^{n/\log n} k^3 n^{2k} \cdot \exp(-\Omega(k \log^6 n)) \\ & \leq \sum_{k=1}^{n/\log n} \exp(-\Omega(k \log^6 n)) \leq \exp(-\Omega(\log^6 n)). \end{aligned}$$

□

3.4.3 Moderate sets: $n/\log n < |S| \leq n - n/\log n$

Claim 3.8. For $n/\log n < k \leq n - n/\log n$ and $\ell \geq n - k + 1$, $\alpha(n, k, \ell) = \Omega(n^4/\log^2 n)$.

Proof. From the formula, $\alpha(n, k, \ell) \geq k \cdot \frac{(n-k+1)(n+k-2)}{2} \cdot (n - \frac{k+1}{2})$. In the moderate regime each factor is at least $n/\log n$, so $\alpha(n, k, \ell) \geq \frac{n}{\log n} \cdot \frac{n}{\log n} \cdot n \cdot \frac{n}{2} = \Omega(n^4/\log^2 n)$.

□

Theorem 3.9 (Hall for Moderate Sets). With probability at least $1 - \exp(-\Omega(n \log^4 n))$, Hall's condition holds for every S with $n/\log n < |S| \leq n - n/\log n$.

Proof. By union bound and Theorem 3.8:

$$\begin{aligned} & \Pr \left[\exists S : \frac{n}{\log n} < |S| \leq n - \frac{n}{\log n}, |N(S)| < |S| \right] \\ & \leq \sum_k \sum_{\ell \geq n-k+1} \binom{n}{k} \binom{n}{\ell} \exp(-p \alpha(n, k, \ell)) \\ & \leq 4^n \cdot \text{poly}(n) \cdot \exp \left(-\frac{\log^6 n}{n^3} \cdot \Omega \left(\frac{n^4}{\log^2 n} \right) \right) \\ & = 4^n \cdot \text{poly}(n) \cdot \exp(-\Omega(n \log^4 n)) = \exp(-\Omega(n \log^4 n)). \end{aligned}$$

□

3.4.4 Near-full sets: $n - n/\log n < |S| \leq n$

Theorem 3.10 (Hall for Near-Full Sets). *With probability at least $1 - \exp(-\Omega(\log^6 n))$, Hall's condition holds for every S with $n - n/\log n < |S| \leq n$.*

Proof. For such S , the complement $\bar{S} = V_1 \setminus S$ satisfies $|\bar{S}| < n/\log n$. By the symmetry of the bipartite construction, Hall's condition for S follows from Hall's condition for $\bar{S}$: if $|N(\bar{S})| \geq |\bar{S}|$, then the vertices in $V_2 \setminus N(\bar{S})$ can only be matched to S , which has more than enough size. Since $|\bar{S}|$ falls in the small-sets regime, Theorem 3.7 applies and gives the same bound $\exp(-\Omega(\log^6 n))$. $\square$

Taking a union bound over all three regimes gives the following.

Theorem 3.11 (Perfect Matching Exists). *With probability at least $1 - \exp(-\Omega(\log^6 n))$, the sparse graph G has a perfect matching, and hence $\text{Det}(X_G) \neq 0$.*

We also need a version that survives after deleting the rows and columns involved in the second-round sampling.

Theorem 3.12 (Perfect Matching After Deletion). *Let $m = \Theta(n^{1/3})$, and let $S_L \subseteq V_1$, $S_R \subseteq V_2$ with $|S_L| = |S_R| = 2m$. With probability at least $1 - \exp(-\Omega(\log^6 n))$, the restricted graph $G_{\text{rest}} = G|_{V_1 \setminus S_L, V_2 \setminus S_R}$ has a perfect matching.*

Proof. Condition on the deletion pattern $\mathcal{D} = (S_L, S_R)$. For a subset $T \subseteq V_1 \setminus S_L$, the neighbourhood in G_{rest} is $N_{G_{\text{rest}}}(T) = N_G(T) \setminus S_R$, so Hall's condition in G_{rest} fails for T only if $|N_G(T) \cap S_R| > |N_G(T)| - |T|$.

Our Hall analysis established neighbourhood expansion with substantial margins. For small T (size at most $n/\log n$), the bound from Theorem 3.7 is $\exp(-\Omega(\log^6 n))$; for moderate T , the bound from Theorem 3.9 is $\exp(-\Omega(n \log^4 n))$; for near-full T , Theorem 3.10 applies. Since $|S_R| = 2m = \Theta(n^{1/3}) = o(n)$, removing these $2m$ vertices cannot violate the expansion: the margins from the concentration bounds absorb the loss. Concretely, invoking Theorem 3.11 on the $(n - 2m) \times (n - 2m)$ subgraph induced by $V_1 \setminus S_L$ and $V_2 \setminus S_R$ (and observing that all relevant parameters scale uniformly) gives the bound.

By the law of total expectation over $\mathcal{D}$, the result follows uniformly over all deletion patterns. $\square$

3.5 The Sparse Matrix

Define the $n \times n$ sparse symbolic matrix X_G by $(X_G)_{i,j} = x_{i,j}$ if $(i, j) \in E_G$ and $(X_G)_{i,j} = 0$ otherwise.

The three properties required for the lower bound argument are now in hand:

1. **Sparsity:** $\Theta(\log^6 n)$ nonzero entries per row and per column (Theorem 3.3).
2. **Nonvanishing:** $\text{Det}(X_G) \neq 0$ via perfect matching (Theorem 3.11).
3. **Diagonal imbalance:** For any $S \subseteq E_G$ with $|S| \geq |E_G|^{0.75}$, at least $(1 - \delta)|S|/2$ rectangles exhibit diagonal imbalance under a random vertex permutation with high probability (Theorem 3.5).

Property (3) is the geometric foundation: rectangle sampling creates an asymmetric structure that, combined with random permutations, will force factor imbalance in the formula decomposition. The details are carried out in Chapter 4.

Chapter 4

Lower Bounds Against Multilinear Formulas

Our polynomial of interest is $\text{Det}(X_G)$, defined over the variable set E_G . Throughout this chapter, $N = |E_G| = \Theta(n \log^6 n)$ denotes the total variable count.

Remark 4.1 Because $N = \Theta(n \log^6 n)$, any quantity expressed as $\Theta(\log N)$ or $\Theta(\log n)$ are asymptotically the same up to constant factors. We use whichever is more convenient.

Lemma 4.2 (Large Factors in the Decomposition). *Under the decomposition of Theorem 2.11 applied to a polynomial over $N = \Theta(n \log^6 n)$ variables, the number of levels j for which $|X_{ij}| \geq N^{0.75}$ is at least $\Theta(\log n)$.*

Proof. The condition $|X_{ij}| \geq N^{0.75}$ requires $(1/3)^j N \geq N^{0.75}$, i.e. $(1/3)^j \geq N^{-0.25}$. Taking logarithms gives $j \leq \frac{0.25 \log N}{\log 3} = \Theta(\log N) = \Theta(\log n)$. Since the total number of levels is $t = \Theta(\log N) = \Theta(\log n)$ and all levels $j \leq \Theta(\log n)$ satisfy the condition, there are $\Theta(\log n)$ such levels. $\square$

4.1 Second-Round Sampling and Concentration

The random restriction operates in a second round: we sample m rectangles from the collection $\mathcal{R}$ built in Chapter 3 and use them to define the (Y, Z) partition.

Definition 4.3 (Sampling Distributions). *We consider two distributions over m -subsets of $\mathcal{R}$:*

- μ^* : sample m rectangles independently and uniformly with replacement.
- μ : sample m rectangles uniformly at random, conditioned on no two sharing a row or column (vertex disjoint).

For $S \subseteq E_G$ and ordering σ , define $Y_S(\sigma, \mathbf{R}) = |\{i \in [m] : R_i \in \mathcal{R}_{S, \text{diag}}(\sigma)\}|$.

In the lower bound argument we use distribution μ (vertex disjoint rectangles), which ensures each variable belongs to at most one sampled rectangle. Two things follow from this: the remaining graph G_{rest} after removing the $2m$ rows and columns retains a perfect matching (Theorem 3.12), and the (Y, Z) partition is clean, with no variable double-counted. The analysis of concentration bounds is however cleaner under μ^* . The statistical distance between μ and μ^* is $o(1)$ for our parameter regime (with $m = \Theta(n^{1/3})$ samples from a pool of size $|\mathcal{R}| = \Theta(n \log^6 n)$, the probability that any two sampled rectangles share a vertex under μ^* is at most $m^2/|\mathcal{R}|^{1/2} = o(1)$ since $m^2 = o(n)$). So any event of probability q under μ^* also holds with probability $q - o(1)$ under μ .

Theorem 4.4 (Concentration of Diagonal Samples). *Let $S \subseteq E_G$ with $|S| \geq N^{0.75}$. Sample $m = \Theta(n^{1/3})$ rectangles under μ^* and let σ be a uniformly random permutation. With probability at least $1 - \exp(-\Omega(n^{1/12} \log^{-1.5} n))$ over both σ and $\mathbf{R}$,*

$$Y_S(\sigma, \mathbf{R}) \geq (1 - \delta') \cdot \frac{m|S|}{2|\mathcal{R}|}$$

for any constant $\delta' > 0$.

Proof. Condition on σ first. For fixed σ , the rectangles $R_1, \dots, R_m$ are sampled independently, so $Y_S \mid \sigma$ is a sum of m independent Bernoulli variables each with mean $|\mathcal{R}_{S, \text{diag}}(\sigma)|/|\mathcal{R}|$. By Theorem 2.12, for any $\delta_1 > 0$,

$$\Pr[Y_S < (1 - \delta_1)\mathbb{E}[Y_S \mid \sigma] \mid \sigma] \leq \exp\left(-\frac{\delta_1^2 \mathbb{E}[Y_S \mid \sigma]}{2}\right).$$

Averaging over σ :

$$\mathbb{E}_{\sigma, \mathbf{R}}[Y_S] = m \cdot \frac{\mathbb{E}_{\sigma}[|\mathcal{R}_{S, \text{diag}}(\sigma)|]}{|\mathcal{R}|} \geq \frac{m|S|}{2|\mathcal{R}|},$$

where the last step uses Theorem 3.5. Let $\mathcal{E}_1$ be the event that $\mathbb{E}[Y_S \mid \sigma] \geq (1 - \delta_2)\mathbb{E}[Y_S]$, and $\mathcal{E}_2$ the event that $Y_S \geq (1 - \delta_1)\mathbb{E}[Y_S \mid \sigma]$. On $\mathcal{E}_1 \cap \mathcal{E}_2$ we get $Y_S \geq (1 - \delta')\mathbb{E}[Y_S]$ with $\delta' = \delta_1 + \delta_2 - \delta_1\delta_2$.

From Theorem 3.5, $\Pr[\mathcal{E}_1^c] \leq 2\exp(-\Omega(n^{0.5} \log^{-3} n))$. For $\Pr[\mathcal{E}_2^c \mid \mathcal{E}_1]$: on $\mathcal{E}_1$,

$$\mathbb{E}[Y_S \mid \sigma] \geq (1 - \delta_2) \cdot \frac{m|S|}{2|\mathcal{R}|} = \Theta\left(\frac{n^{1/3} \cdot n^{0.75} \log^{4.5} n}{n \log^6 n}\right) = \Theta(n^{1/12} \log^{-1.5} n).$$

The Chernoff bound then gives $\Pr[\mathcal{E}_2^c \mid \sigma] \leq \exp(-\Omega(n^{1/12} \log^{-1.5} n))$ uniformly

over $\sigma \in \mathcal{E}_1$. Combining these two bounds:

$$\begin{aligned} \Pr[(\mathcal{E}_1 \cap \mathcal{E}_2)^c] &\leq 2 \exp(-\Omega(n^{0.5} \log^{-3} n)) + \exp(-\Omega(n^{1/12} \log^{-1.5} n)) \\ &= \exp(-\Omega(n^{1/12} \log^{-1.5} n)). \end{aligned} \quad \square$$

Remark 4.5 The content of Theorem 4.4 is that $Y_S = \Omega(n^{1/12} \log^{-1.5} n)$ with high probability whenever $|S| \geq N^{0.75}$. Informally: any large set of variables has many rectangle-witnesses to diagonal imbalance in the sample.

4.2 The Random Restriction

The random restriction defines an equi-partition of the original variables into active sets Y, Z (which feed into Raz's measure) and frozen constants. This argument follows the structure of [Raz09, Section 6], adapted to the sparse setting.

Step 1: Sample rectangles. Draw $\mathbf{R} = (R_1, \dots, R_m)$ with $m = \Theta(n^{1/3})$ from distribution μ . Each R_i involves rows $\{r_1^i, r_2^i\}$ and columns $\{c_1^i, c_2^i\}$. Set $S_L = \bigcup_i \{r_1^i, r_2^i\}$ and $S_R = \bigcup_i \{c_1^i, c_2^i\}$; since the rectangles are vertex disjoint, $|S_L| = |S_R| = 2m$.

Step 2: Assign variables in sampled rectangles. For each R_i , independently choose one of two diagonal configurations with equal probability:

- *Config 1:* $x_{r_1^i, c_1^i} \mapsto y_i$, $x_{r_2^i, c_2^i} \mapsto z_i$, $x_{r_1^i, c_2^i} = x_{r_2^i, c_1^i} = 1$.
- *Config 2:* $x_{r_1^i, c_1^i} \mapsto z_i$, $x_{r_2^i, c_2^i} \mapsto y_i$, $x_{r_1^i, c_2^i} = x_{r_2^i, c_1^i} = 1$.

Step 3: Freeze remaining variables. By Theorem 3.12, the residual graph G_{rest} on $V_1 \setminus S_L$ and $V_2 \setminus S_R$ has a perfect matching M_{rest} with high probability. Fix any such matching, and set $x_e = 1$ for $e \in M_{\text{rest}}$ and $x_e = 0$ for the remaining edges of G_{rest} .

After the restriction, the active variables are $Y = \{y_1, \dots, y_m\}$ and $Z = \{z_1, \dots, z_m\}$. For a factor f_{ij} over variable set X_{ij} , write $Y_{ij} = Y \cap X_{ij}$ and $Z_{ij} = Z \cap X_{ij}$.

4.3 Balance Analysis

Lemma 4.6 (Binomial Distribution of $|Y_{ij}|$). *Consider a factor f_{ij} with $|X_{ij}| \geq N^{0.75}$. After the random restriction with rectangles from μ (or μ^*), the count $|Y_{ij}|$*

is distributed as $\text{Binomial}(Y_S, 1/2)$ where $Y_S = \Omega(n^{1/12} \log^{-1.5} n)$ with probability at least $1 - \exp(-\Omega(n^{1/12} \log^{-1.5} n))$.

Proof. By Theorem 4.4, with the stated probability there are at least $Y_S = \Omega(n^{1/12} \log^{-1.5} n)$ sampled rectangles contributing a diagonally imbalanced pair to X_{ij} . For each such rectangle R_ℓ , exactly one corner falls in X_{ij} on the active (y or z) diagonal. The configuration choice determines which of y_ℓ or z_ℓ goes to Y_{ij} , and this choice is made independently and uniformly for each rectangle. Hence $|Y_{ij}|$ is a sum of Y_S independent Bernoulli(1/2) variables. $\square$

Proposition 4.7 (Balance is Unlikely for Large Factors). *For a factor f_{ij} with $|X_{ij}| \geq N^{0.75}$ and $k = n^{1/32}$, $\Pr[f_{ij} \text{ is } k\text{-balanced}] = O(n^{-1/96} \log^{0.75} n)$.*

Proof. Let $\mathcal{G}_{ij}$ be the event that at least $Y_S = \Omega(n^{1/12} \log^{-1.5} n)$ sampled rectangles are diagonally imbalanced relative to X_{ij} . By Theorem 4.4, $\Pr[\mathcal{G}_{ij}] \geq 1 - \exp(-\Omega(n^{1/12} \log^{-1.5} n))$.

On $\mathcal{G}_{ij}$, by Theorem 4.6, $|Y_{ij}| \sim \text{Binomial}(Y_S, 1/2)$. The factor is k -balanced iff $||Y_{ij}| - Y_S/2| < k$. By the normal approximation to the binomial,

$$\Pr[k\text{-balanced} \mid \mathcal{G}_{ij}] = O\left(\frac{k}{\sqrt{Y_S}}\right) = O\left(\frac{n^{1/32}}{\sqrt{n^{1/12} \log^{-1.5} n}}\right) = O(n^{-1/96} \log^{0.75} n).$$

By the law of total probability, $\Pr[k\text{-balanced}] \leq O(n^{-1/96} \log^{0.75} n) + \exp(-\Omega(n^{1/12} \log^{-1.5} n)) = O(n^{-1/96} \log^{0.75} n)$. $\square$

4.4 Read-2 Structure of the Balance Events

Lemma 4.8 (Read-2 Family). *Condition on the sampled rectangles $\mathbf{R}$. The balance events for the factors in any single summand of the formula decomposition form a read-2 family with respect to the configuration choices $C_1, \dots, C_m$.*

Proof. Given $\mathbf{R}$, the remaining randomness is the m independent uniform choices $C_i \in \{1, 2\}$. Rectangle R_i contributes exactly two active variables: y_i and z_i . Within a single summand $f_i = \prod_j f_{ij}$, the variable sets of the factors are pairwise disjoint. So each active variable belongs to at most one factor. Since R_i contributes at most two active variables, C_i influences the balance of at most two factors. $\square$

4.5 Many Factors are Simultaneously Imbalanced

Theorem 4.9 (Simultaneous Imbalance). *Let $\mathcal{L}$ be the set of $r = \Theta(\log n)$ levels j with $|X_{ij}| \geq N^{0.75}$ in a fixed summand f_i . With probability at least*

$1 - \exp(-\Omega(\log^2 n))$ over the rectangle sample and configuration choices, at least $r/2$ factors in $\mathcal{L}$ are k -imbalanced for $k = n^{1/32}$.

Proof. For $\ell \in \mathcal{L}$, let $B_\ell = \mathbf{1}[f_{i\ell} \text{ is } k\text{-balanced}]$. Let $\mathcal{G}$ be the event that every factor in $\mathcal{L}$ receives at least $\Omega(n^{1/12} \log^{-1.5} n)$ imbalanced rectangle-witnesses. By Theorem 4.4, $\Pr[\mathcal{G}^c] \leq \exp(-\Omega(n^{1/12} \log^{-1.5} n))$.

On $\mathcal{G}$, by Theorem 4.8 the events $\{B_\ell\}_{\ell \in \mathcal{L}}$ form a read-2 family over the configurations. By Theorem 4.7, each satisfies $\Pr[B_\ell = 1 \mid \mathbf{R}] = O(n^{-1/96} \log^{0.75} n) =: p$. For $q = 1/2$ and $p \ll 1/2$, the divergence satisfies $D(1/2 \parallel p) = \frac{1}{2} \log \frac{1}{2p} = \Theta(\log n)$. By Theorem 2.15 with $k = 2$ (read-2 family):

$$\Pr\left[\sum_{\ell \in \mathcal{L}} B_\ell \geq \frac{r}{2} \mid \mathbf{R} \in \mathcal{G}\right] \leq \exp\left(-\frac{r}{2} \cdot D(1/2 \parallel p)\right) = \exp(-\Omega(\log^2 n)).$$

Adding the probability of $\mathcal{G}^c$ and noting $\exp(-\Omega(n^{1/12} \log^{-1.5} n)) \ll \exp(-\Omega(\log^2 n))$ gives the result. $\square$

4.6 The Lower Bound

Theorem 4.10 (Lower Bound for Sparse Determinants). *Any multilinear formula computing $\text{Det}(X_G)$ has size $s = n^{\Omega(\log n)}$.*

Proof. Suppose a multilinear formula F of size $s = n^{O(\log n)}$ computes $\text{Det}(X_G)$. By Theorem 2.11, F decomposes as $\sum_{i=1}^{s+1} f_i$ with each $f_i = \prod_{j=1}^t f_{ij}$, $t = \Theta(\log n)$. By Theorem 4.2, each summand contains $r = \Theta(\log n)$ levels j with $|X_{ij}| \geq N^{0.75}$.

Fix a summand f_i . By Theorem 4.9, the random restriction causes at least $\ell = r/2 = \Theta(\log n)$ of the large factors to be k -imbalanced (for $k = n^{1/32}$) with probability at least $1 - \exp(-\Omega(\log^2 n))$. On this event, Theorem 2.10 gives

$$\mu_\sigma(f_i) \leq 2^m \cdot 2^{-\ell k} = 2^m \cdot 2^{-\Omega(\log n) \cdot n^{1/32}} = 2^m \cdot 2^{-\Omega(n^{1/32} \log n)}.$$

Union-bounding over all $s+1$ summands: if $s = n^{O(\log n)}$, then with probability at least $1 - (s+1) \exp(-\Omega(\log^2 n)) = 1 - o(1)$, every summand simultaneously satisfies the rank bound. By subadditivity of rank,

$$\mu_\sigma(F) \leq \sum_{i=1}^{s+1} \mu_\sigma(f_i) \leq (s+1) \cdot 2^m \cdot 2^{-\Omega(n^{1/32} \log n)}.$$

For $s = n^{O(\log n)}$ and $m = \Theta(n^{1/3})$, since $2^{\Omega(n^{1/32} \log n)} \gg n^{O(\log n)}$ for large n , the right-hand side is $\ll 2^m$.

On the other hand, $\text{Det}(X_G)$ maintains full rank. By Theorem 3.12, G_{rest} has a perfect matching M_{rest} with probability $1 - \exp(-\Omega(\log^6 n))$. Under the restriction, for each sampled rectangle R_i the two active corners become (y_i, z_i) or (z_i, y_i) and the other diagonal is set to 1; the remaining edges are assigned 0 or 1 via M_{rest} . The restricted determinant then expands as

$$\text{Det}(X_G)|_{\text{restricted}} = \prod_{i=1}^m (y_i z_i - 1).$$

This polynomial contains all 2^m monomials over $Y = \{y_1, \dots, y_m\}$ (each paired with the complementary monomial in Z), so the coefficient matrix M_{Det} is a full-rank $2^m \times 2^m$ matrix. Therefore $\mu_\sigma(\text{Det}(X_G)) = 2^m$.

This contradicts $\mu_\sigma(F) \ll 2^m$. Hence $s = n^{\Omega(\log n)}$. □

Chapter 5

Discussion

Why structured sparse matrices do not immediately work. A natural first thought is to use a deterministic sparse matrix: a band matrix, a block diagonal structure, or a matrix supported on a constant-degree expander. Deterministic constructions are cleaner and fully explicit, which is appealing.

The obstacle is that Raz’s argument needs a source of diagonal imbalance that no adversarial formula can avoid. In the dense setting, random permutations of the n^2 variables produce imbalance for any large factor, essentially for counting reasons. In the sparse setting, a formula could potentially align its variable sets with the predetermined sparsity pattern. For instance, if the matrix has a block diagonal structure with n blocks of size $\sqrt{n} \times \sqrt{n}$, a formula of the right depth could arrange its factors to intersect exactly one block each, sidestepping imbalance entirely.

Rectangle sampling avoids this by making the random permutation over vertices (not variables) the source of imbalance. Any large set $S \subseteq E_G$, however adversarially chosen, sees roughly half its rectangles exhibit diagonal imbalance under a random permutation (Theorem 3.5). Since the formula has no knowledge of the permutation in advance, this imbalance cannot be avoided.

Two sources of randomness. Two separate random choices appear in the argument, and it helps to keep them apart.

The graph G itself is built by sampling: each rectangle is included at probability $p = \log^6 n / n^3$, independently. This step produces the sparse support and is the focus of the Hall analysis in Chapter 3.

Choosing the permutation $\sigma = (\sigma_L, \sigma_R)$ is a separate random step, and it serves a different purpose. Even after G is fixed, a fresh uniformly random permutation is needed to establish diagonal imbalance. Without it, every rectangle’s corners are in fixed positions and an adversarial variable set could concentrate its mass on one diagonal everywhere, evading imbalance. The permutation removes this

possibility: each variable lands at each corner position with equal probability, so no large set can avoid having roughly half its rectangles imbalanced.

Raz’s original argument [Raz09] embeds m disjoint 2×2 blocks at random positions along a generalised diagonal of the dense matrix. Our two-stage process plays the same structural role: the graph gives the rectangles, the permutation gives the imbalance.

The choice of parameter $\log^6 n$. The sampling probability $p = \log^6 n / n^3$ is chosen so that each vertex has degree $\Theta(\log^6 n)$. The exponent 6 is not the tightest achievable.

Three constraints in the concentration arguments push on the parameter:

- The McDiarmid bound (Theorem 3.5) requires the variation from changing one vertex position to be manageable, which is controlled by the maximum degree $O(\log^6 n)$.
- The Hall analysis for small sets (Theorem 3.7) requires neighbourhood expansion at rate $k \log^6 n$ for a set of size k , which degree $\Theta(\log^6 n)$ exactly provides.
- The second-round sampling needs $m = \Theta(n^{1/3})$ vertex disjoint rectangles, which exist once $|\mathcal{R}| = \Theta(n \log^6 n) \gg n^{1/3}$.

The exponent 6 gives comfortable margins across all three. The proof goes through with any slower-growing function of n in the degree, though the constants would need retracing.

VBP-hardness of sparse determinants. Sparse determinants are VBP-hard, inheriting this from the general determinant via a direct reduction. The Mahajan-Vinay characterisation [MV97] expresses the determinant of any $n \times n$ matrix as a polynomial-size ABP of width n . To encode this ABP as a sparse determinant, replace each high-degree node by a binary tree of intermediate nodes so that the support graph has constant maximum degree. The width grows by at most a logarithmic factor and the size by a polynomial. Since this is a polynomial-size projection, sparse determinants are VBP-hard under p-projections. Any formula of size $n^{o(\log n)}$ for sparse determinants would therefore collapse VF to VBP, which would be a much stronger separation than what is proved here and is widely believed to be false.

Conclusion. Put plainly: density does not drive the multilinear hardness of the determinant. Stripping out all but $\Theta(n \log^6 n)$ of the n^2 variables does not weaken

the lower bound. What survives in the sparse graph is enough rectangle geometry that any large factor in a multilinear formula cannot escape imbalance under a random permutation.

What actually matters, to us, is what kind of property does drive hardness. The point is that rectangles are everywhere in G and unavoidable for any large variable set. This looks like a form of expansion. The variable set cannot concentrate in a way that evades imbalanced rectangles.

Whether this extends to even sparser graphs or to other polynomial families is the natural next question. For the determinant, the limiting factor is needing a perfect matching. For other polynomials, analogous structures might arise differently. We hope this analysis points toward lower bounds for polynomials that have so far resisted Raz's technique.

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